Class 10 Chemistry - Solved Exercises (PCTB) - Chapters 9-16

Chapter 9: Chemical Equilibrium

Short Questions

1. What are irreversible reactions? Give a few characteristics of them.

Irreversible reactions are reactions that proceed only in the forward direction and go to completion, so the reactants are fully converted into products. Their characteristics are: they occur in one direction only, they do not attain equilibrium, the products cannot be converted back into reactants under the same conditions, and examples are burning of fuel and rusting of iron.

2. Define chemical equilibrium state.

Chemical equilibrium is the state of a reversible reaction in which the rate of the forward reaction becomes equal to the rate of the reverse reaction. At this state the concentrations of reactants and products stop changing with time, although both reactions continue to occur at equal rates.

3. Give the characteristics of reversible reaction.

A reversible reaction can proceed in both forward and reverse directions, it does not go to completion, and it can attain a state of dynamic equilibrium. It is represented by a double arrow, and its direction can be

changed by changing conditions such as concentration, temperature or pressure.

4. How is dynamic equilibrium established?

Dynamic equilibrium is established when a reversible reaction is carried out in a closed vessel. At first the forward reaction is fast and the reverse reaction is slow, but as products build up the reverse rate increases and the forward rate decreases. When the two rates become equal, dynamic equilibrium is established and concentrations remain constant.

5. Why at equilibrium state reaction does not stop?

At equilibrium the reaction does not stop because both the forward and reverse reactions continue to take place. They occur at exactly equal rates, so the amount of reactants converted into products in a given time equals the amount of products converted back into reactants. The net change is therefore zero, which is why it is called dynamic equilibrium.

6. Why is equilibrium state attainable from either way?

Equilibrium can be reached either by starting with reactants or by starting with products, because the same equilibrium state is determined only by the conditions (temperature and concentrations). Whether the reaction moves forward or backward, the rates adjust until they become equal, giving the same equilibrium concentrations in both cases.

7. What is relationship between active mass and rate of reaction?

Active mass means the molar concentration of a substance, in mol/dm^3. According to the law of mass action, the rate of a reaction is directly proportional to the product of the active masses of the reactants,

each raised to the power of its coefficient in the balanced equation. Increasing concentration therefore increases the reaction rate.

8. Derive equilibrium constant expression for the synthesis of ammonia from nitrogen and hydrogen.

The synthesis of ammonia is: N2 + 3H2 ⇌ 2NH3. The rate of the forward reaction is proportional to [N2][H2]^3 and the rate of the reverse reaction is proportional to [NH3]^2. At equilibrium the two rates are equal, so Kc = [NH3]^2 / ([N2][H2]^3). This is the equilibrium constant expression for ammonia synthesis.

9. Write the equilibrium constant expression for the following reactions (reactions printed as figures in the book).

The general method is: write the balanced equation, then Kc = (product concentrations raised to their coefficients) divided by (reactant concentrations raised to their coefficients). For example, for the synthesis of ammonia, N2 + 3H2 ⇌ 2NH3, Kc = [NH3]^2 / ([N2][H2]^3). Apply the same rule to each reaction printed in the book, using the coefficients shown in its equation.

10. How can direction of a reaction be predicted?

The direction of a reaction is predicted by comparing the reaction quotient Qc with the equilibrium constant Kc. Qc is calculated from the current concentrations in the same form as Kc. If Qc is less than Kc, the reaction moves forward; if Qc is greater than Kc, it moves backward; if Qc equals Kc, the reaction is already at equilibrium.

11. How can you know that a reaction has achieved an equilibrium state?

A reaction has reached equilibrium when the concentrations of reactants and products stop changing with time, and when observable properties such as colour, pressure or density become constant. At this point the forward and reverse rates are equal, although both reactions are still occurring.

12. What are the characteristics of a reaction that establishes equilibrium state at once?

A reaction that reaches equilibrium at once is usually a fast ionic reaction in solution, such as a neutralization or precipitation reaction, where ions meet and react almost instantly. Such reactions have very high rates in both directions, so the balance between forward and reverse rates is established immediately.

13. If reaction quotient Qc of a reaction is more than Kc, what will be the direction of the reaction?

If Qc is greater than Kc, there is too much product (or too little reactant) compared with equilibrium. The reaction will therefore move in the reverse direction, converting products back into reactants, until Qc falls to equal Kc and equilibrium is restored.

14. An industry was established based upon a reversible reaction. It failed to achieve products on commercial level. Can you point out the basic reasons of its failure being a chemist?

The failure may be because the equilibrium lies far to the left, so the yield of product is very low, or because the reaction is too slow at the chosen conditions. Other reasons are: unsuitable temperature or pressure, absence of a catalyst, an open system from which a reactant or

product escapes, or failure to remove the product continuously to shift the equilibrium forward.

Extensive Questions

1. Describe a reversible reaction with the help of an example and graph.

A reversible reaction is one that can proceed in both directions, for example the synthesis of ammonia: $N_2 + 3H_2 \rightleftharpoons 2NH_3$. On a graph of concentration against time, the reactant concentrations fall and the product concentration rises at first, then all curves become horizontal. The flat parts of the graph show that concentrations have become constant, which is the equilibrium state.

2. Write down the macroscopic characteristics of dynamic equilibrium.

The macroscopic characteristics of dynamic equilibrium are: the concentrations of reactants and products remain constant, the observable properties such as colour, pressure and density do not change, the reaction can be approached from either direction, it occurs only in a closed system, and a catalyst does not change the equilibrium position but helps to reach it faster.

3. State the law of Mass Action and derive the expression for equilibrium constant for a general reaction.

The law of mass action states that the rate of a chemical reaction is directly proportional to the product of the active masses (molar concentrations) of the reactants, each raised to the power of its coefficient. For a general reaction $aA + bB \rightleftharpoons cC + dD$, forward rate =

kf[A]^a[B]^b and reverse rate = kr[C]^c[D]^d. At equilibrium these rates are equal, so kf/kr = ([C]^c[D]^d)/([A]^a[B]^b) = Kc, the equilibrium constant.

4. What is the importance of equilibrium constant?

The equilibrium constant Kc tells the extent of a reaction: a large Kc means the equilibrium lies far to the right and the yield of products is high, while a small Kc means the yield is low. It also predicts the direction of a reaction by comparison with Qc, and helps industries choose conditions (temperature, pressure, concentration) that give the maximum yield of product.

Numericals

1. For the decomposition of dinitrogen oxide (N2O) into nitrogen and oxygen, the equilibrium concentrations of N2O, N2 and O2 are 1.1, 3.90 and 1.95 mol/dm^3 respectively. Find Kc for this reaction. (Reaction: 2N2O ⇌ 2N2 + O2)

Step 1: Write the Kc expression: Kc = [N2]^2 x [O2] / [N2O]^2.

Step 2: Substitute: Kc = (3.90)^2 x 1.95 / (1.1)^2 = 15.21 x 1.95 / 1.21.

Step 3: Kc = 29.66 / 1.21 = 24.5.

Final answer: Kc = 24.5 mol/dm^3.

2. Hydrogen iodide decomposes to form hydrogen and iodine. If the equilibrium concentration of HI is 0.078 mol/dm^3, and H2 and I2 are each 0.011 mol/dm^3, calculate the equilibrium constant value for this

reversible reaction. (Reaction: 2HI ⇌ H2 + I2)

Step 1: Kc = [H2] x [I2] / [HI]^2.

Step 2: Kc = 0.011 x 0.011 / (0.078)^2 = 0.000121 / 0.006084.

Step 3: Kc = 0.0199, about 0.02.

Final answer: Kc = 0.02 (no units, as the units cancel).

3. For the fixation of nitrogen, N2 + O2 ⇌ 2NO, at 1500 K the Kc is 1.1 x 10^-5. If equilibrium concentrations of nitrogen and oxygen are 1.7 x 10^-3 and 6.4 x 10^-3 mol/dm^3 respectively, how much NO is formed?

Step 1: Kc = [NO]^2 / ([N2] x [O2]), so [NO]^2 = Kc x [N2] x [O2].

Step 2: [NO]^2 = 1.1 x 10^-5 x 1.7 x 10^-3 x 6.4 x 10^-3 = 1.197 x 10^-10.

Step 3: [NO] = square root (1.197 x 10^-10) = 1.09 x 10^-5 mol/dm^3.

Final answer: NO formed = 1.09 x 10^-5 mol/dm^3.

4. When nitrogen reacts with hydrogen to form ammonia, the equilibrium mixture contains 0.31 mol/dm^3 of nitrogen and 0.50 mol/dm^3 of hydrogen. If Kc is 0.50 mol^-2 dm^6, what is the equilibrium concentration of ammonia? (Reaction: N2 + 3H2 ⇌ 2NH3)

Step 1: Kc = [NH3]^2 / ([N2] x [H2]^3), so [NH3]^2 = Kc x [N2] x [H2]^3.

Step 2: [NH3]^2 = 0.50 x 0.31 x (0.50)^3 = 0.50 x 0.31 x 0.125 = 0.019375.

Step 3: [NH3] = square root (0.019375) = 0.139 mol/dm^3.

Final answer: Equilibrium concentration of ammonia = 0.139 mol/dm^3.

Chapter 10: Acids, Bases and Salts

Short Questions

1. Name three common household substances having (a) pH greater than 7 (b) pH less than 7 (c) pH equal to 7.

(a) pH greater than 7: soap solution, washing soda solution, and baking soda solution. (b) pH less than 7: vinegar, lemon juice, and tomato juice. (c) pH equal to 7: pure water, common salt solution, and sugar solution.

2. Define a base and explain that all alkalies are bases, but all bases are not alkalies.

A base is a substance that neutralizes an acid to form a salt and water, and accepts protons or donates hydroxide ions. An alkali is a base that dissolves in water to give hydroxide ions. All alkalies are bases because they neutralize acids, but all bases are not alkalies because some bases, such as copper oxide or calcium carbonate, do not dissolve in water and so are not alkalies.

3. Define Bronsted-Lowry base and explain with an example that water is a Bronsted-Lowry base.

A Bronsted-Lowry base is a substance that accepts a proton (H+ ion) from an acid. Water acts as a Bronsted-Lowry base when it reacts with hydrogen chloride: HCl + H2O → H3O+ + Cl-. Here water accepts a proton from HCl to form the hydronium ion, so water behaves as a base.

4. How can you justify that Bronsted-Lowry concept of acid and base is applicable to non-aqueous solutions?

The Bronsted-Lowry concept is based on proton transfer, not on water or hydroxide ions, so it works in any solvent where proton transfer can occur. For example, in liquid ammonia, ammonium chloride acts as an acid by donating a proton and amide ion acts as a base by accepting it, even though no water is present.

5. Which kind of bond is formed between Lewis acid and a base?

A coordinate covalent bond (dative bond) is formed between a Lewis acid and a Lewis base. The base donates a lone pair of electrons to the acid, which accepts it into a vacant orbital, and the shared pair forms the bond between them.

6. Why does H+ ion act as a Lewis acid?

A Lewis acid is an electron-pair acceptor. The H+ ion has no electrons and has a vacant orbital, so it readily accepts a lone pair of electrons from a base, for example from water to form H3O+. It therefore behaves as a Lewis acid.

7. Name two acids used in the manufacture of fertilizers.

Two acids used in fertilizer manufacture are sulphuric acid, used to make ammonium sulphate and superphosphate, and nitric acid, used to make ammonium nitrate and calcium ammonium nitrate.

8. Define pH. What is the pH of pure water?

pH is the negative logarithm of the hydrogen ion concentration, pH = -log[H+]. It measures how acidic or basic a solution is on a scale from 0

to 14. The pH of pure water at 25 degrees C is 7, because [H+] = 10^-7 mol/dm^3, so water is neutral.

9. How many times a solution of pH 1 will be stronger than that of a solution having pH 2?

The pH scale is logarithmic, so each whole unit of pH means a tenfold change in hydrogen ion concentration. A solution of pH 1 has [H+] = 10^-1, while pH 2 has [H+] = 10^-2. The pH 1 solution is therefore 10 times stronger (more acidic) than the pH 2 solution.

10. Define the following: (i) Normal salt (ii) Basic salt.

(i) A normal salt is formed when all the replaceable hydrogen atoms of an acid are replaced by a metal or ammonium ion, for example NaCl and Na2SO4. (ii) A basic salt is formed when only some of the hydroxide ions of a base are replaced by an acid radical, so it still contains a hydroxide group, for example Pb(OH)NO3 and Mg(OH)Cl.

11. Na2SO4 is a neutral salt while NaHSO4 is an acid salt. Justify.

Na2SO4 is formed when both hydrogen atoms of sulphuric acid are replaced by sodium, so it has no replaceable hydrogen left and its solution is neutral. NaHSO4 is formed when only one hydrogen atom is replaced, leaving one acidic hydrogen (HSO4-) that can still be donated, so it behaves as an acid salt.

12. Give a few characteristic properties of salts.

Salts are generally crystalline solids with high melting points, they are formed by neutralization of acids and bases, many dissolve in water to give neutral, acidic or basic solutions, they conduct electricity when

molten or in solution, and they have characteristic colours and tastes (for example copper sulphate is blue).

13. How are the soluble salts recovered from water?

Soluble salts are recovered from water by evaporation or crystallization. The salt solution is heated gently to evaporate the water, or it is concentrated and then cooled so that pure crystals of the salt separate out, which are then filtered and dried.

14. How are the insoluble salts prepared?

Insoluble salts are prepared by precipitation: two soluble salt solutions are mixed so that the insoluble salt forms as a solid precipitate. The precipitate is then filtered off, washed with distilled water to remove soluble impurities, and dried.

15. Why is a salt neutral? Explain with an example.

A salt is neutral when it is formed from a strong acid and a strong base, because neither of its ions reacts with water to produce excess H+ or OH- ions. For example, sodium chloride is formed from strong acid HCl and strong base NaOH, so its solution has pH 7 and is neutral.

16. Name an acid used in the preservation of food.

Acetic acid (vinegar) is used in the preservation of food, for example in pickles and sauces. Citric acid is also used to preserve jams, squashes and soft drinks.

17. Name the acids present in: (i) Vinegar (ii) Ant sting (iii) Citrus fruit (iv) Sour milk.

(i) Vinegar: acetic acid. (ii) Ant sting: formic acid (methanoic acid). (iii) Citrus fruit: citric acid. (iv) Sour milk: lactic acid.

18. How can you justify that Pb(OH)NO3 is a basic salt?

Pb(OH)NO3 still contains a hydroxide (OH) group from the base Pb(OH)2, because only one of the two hydroxide groups has been replaced by the nitrate group. Since it contains both a basic hydroxide part and an acid radical, it is a basic salt.

19. You are in a need of an acidic salt. How can you prepare it?

An acidic salt is prepared by partially neutralizing a polybasic acid with a base, using less base than needed for full neutralization. For example, adding a limited amount of NaOH to sulphuric acid gives sodium hydrogen sulphate (NaHSO4), an acidic salt that still contains a replaceable hydrogen atom.

20. Which salt is used to prepare plaster of Paris?

Calcium sulphate dihydrate (gypsum, CaSO4.2H2O) is heated to about 120 degrees C to remove part of its water of crystallization, giving calcium sulphate hemihydrate (CaSO4.1/2H2O), which is plaster of Paris.

Extensive Questions

1. Define an acid and a base according to Bronsted-Lowry concept and justify with examples that water is an amphoteric compound.

According to Bronsted-Lowry, an acid donates a proton and a base accepts a proton. Water is amphoteric because it can do both: with HCl it

accepts a proton (HCl + H2O → H3O+ + Cl-) and acts as a base, while with ammonia it donates a proton (NH3 + H2O → NH4+ + OH-) and acts as an acid. Water can therefore behave as either an acid or a base depending on the other substance.

2. Explain the Lewis concept of acids and bases.

According to Lewis, an acid is an electron-pair acceptor and a base is an electron-pair donor. When a base donates a lone pair to an acid, a coordinate covalent bond forms. For example, ammonia (base) donates its lone pair to boron trifluoride (acid) to form NH3.BF3. This concept is wider than earlier ones because it also covers reactions with no protons or hydrogen ions.

3. What is auto-ionization of water? How is it used to establish the pH of water?

Auto-ionization (self-ionization) of water is the reaction in which two water molecules transfer a proton between themselves: 2H2O ⇌ H3O+ + OH-. At 25 degrees C the product [H+][OH-] = 10^-14, and since [H+] = [OH-] in pure water, [H+] = 10^-7 mol/dm^3. Therefore pH = -log(10^-7) = 7, establishing that pure water is neutral.

4. Define a salt and give the characteristic properties of salts.

A salt is an ionic compound formed when the hydrogen of an acid is replaced by a metal or ammonium ion, usually by neutralization of an acid with a base. Its properties are: it is a crystalline solid, it has a high melting point, it is often soluble in water, its solution or melt conducts electricity, and it may be neutral, acidic or basic depending on the strength of the acid and base from which it came.

5. Explain with examples how soluble salts are prepared.

Soluble salts are prepared by: (i) neutralization of an acid with a base, for example HCl + NaOH → NaCl + H2O; (ii) reaction of an acid with a metal, for example 2HCl + Zn → ZnCl2 + H2; (iii) reaction of an acid with a metal oxide or carbonate, for example 2HCl + CaCO3 → CaCl2 + CO2 + H2O. The salt is then recovered by evaporation or crystallization.

6. Give the characteristics of an acidic salt.

An acidic salt still contains one or more replaceable hydrogen atoms, so its solution is acidic (pH less than 7). It is formed by partial neutralization of a polybasic acid, for example NaHSO4 and NaHCO3. On heating or on reaction with a base it can lose its remaining hydrogen and change into a normal salt.

7. Give four uses of calcium oxide.

Four uses of calcium oxide (quicklime) are: in the manufacture of cement and mortar; to neutralize acidic soils in agriculture; in the purification of sugar and in water treatment to remove hardness; and in the manufacture of bleaching powder and glass.

8. You are having a strong acid (HNO3) and strong base (NaOH) on mixing: (i) What type of salt will you have? (ii) What type of reaction will it be? (iii) Will it be soluble or insoluble salt? (iv) If it is soluble, how will it be recovered?

(i) A normal, neutral salt, sodium nitrate (NaNO3). (ii) A neutralization reaction: HNO3 + NaOH → NaNO3 + H2O. (iii) It will be a soluble salt, as all nitrates and sodium salts are soluble. (iv) It is recovered by evaporating the water, or by concentrating the solution and cooling it so

that crystals of sodium nitrate separate, which are then filtered and dried.

9. Explain why: (i) HCl forms only one series of salts. (ii) H2SO4 forms two series of salts. (iii) H3PO4 forms three series of salts. Give necessary equations.

The number of series of salts equals the number of replaceable hydrogen atoms in the acid. (i) HCl is monobasic with one hydrogen: HCl + NaOH → NaCl + H2O, giving only one series. (ii) H2SO4 is dibasic: partial neutralization gives NaHSO4, full neutralization gives Na2SO4, giving two series. (iii) H3PO4 is tribasic: it gives NaH2PO4, Na2HPO4 and Na3PO4, giving three series of salts.

10. Classify the following salts as soluble or insoluble salts: (i) Sodium chloride (ii) Silver nitrate (iii) Lead chloride (iv) Copper sulphate (v) Barium sulphate (vi) Ammonium chloride (vii) Sodium carbonate (viii) Calcium carbonate (ix) Ferric chloride (x) Magnesium sulphate.

Soluble: sodium chloride, silver nitrate, copper sulphate, ammonium chloride, sodium carbonate, ferric chloride, magnesium sulphate.

Insoluble (or sparingly soluble): lead chloride (soluble only in hot water), barium sulphate, calcium carbonate. The general rules are: all sodium, potassium, ammonium and nitrate salts are soluble; most sulphates are soluble except barium, lead and calcium sulphates; most carbonates are insoluble except those of sodium, potassium and ammonium.

11. Complete and balance the following equations (equations printed as figures in the book).

Balance each equation by making the number of atoms of every element

equal on both sides, following the method: write the unbalanced equation, count atoms on each side, adjust coefficients starting with the most complex compound, and check that all atoms and charges balance. Apply this to each equation printed in the book, keeping the reactants and products exactly as shown there.

Numericals

1. Calculate the pH and pOH of 0.2 M H2SO4.

Step 1: Sulphuric acid gives two protons; taking both into account, [H+] = 2 x 0.2 = 0.4 mol/dm^3.

Step 2: pH = -log(0.4) = 0.40.

Step 3: pOH = 14 - pH = 14 - 0.40 = 13.60.

Final answer: pH = 0.40, pOH = 13.60 (for the first proton only, pH = 0.70).

2. Calculate the pH of 0.1 M KOH.

Step 1: [OH-] = 0.1 mol/dm^3.

Step 2: pOH = -log(0.1) = 1.

Step 3: pH = 14 - pOH = 14 - 1 = 13.

Final answer: pH = 13.

3. Calculate the pOH of 0.004 M HNO3.

Step 1: [H+] = 0.004 mol/dm^3.

Step 2: pH = -log(0.004) = 2.40.

Step 3: pOH = 14 - 2.40 = 11.60.

Final answer: pOH = 11.60.

4. Complete the following table: (i) 0.15 M HI (ii) 0.040 M KOH (iii) 0.020 M Ba(OH)2 (iv) 0.00030 M HClO4 (v) 0.55 M NaOH (vi) 0.055 M HCl (vii) 0.055 M Ca(OH)2. Give [H+], [OH-], pH and pOH for each.

Step 1: (i) 0.15 M HI: [H+] = 0.15, pH = 0.82, pOH = 13.18, [OH-] = 6.6 x 10^-14.

Step 2: (ii) 0.040 M KOH: [OH-] = 0.040, pOH = 1.40, pH = 12.60, [H+] = 2.5 x 10^-13.

Step 3: (iii) 0.020 M Ba(OH)2: [OH-] = 0.040, pOH = 1.40, pH = 12.60, [H+] = 2.5 x 10^-13.

Step 4: (iv) 0.00030 M HClO4: [H+] = 3.0 x 10^-4, pH = 3.52, pOH = 10.48, [OH-] = 3.3 x 10^-11.

Step 5: (v) 0.55 M NaOH: [OH-] = 0.55, pOH = 0.26, pH = 13.74, [H+] = 1.8 x 10^-14.

Step 6: (vi) 0.055 M HCl: [H+] = 0.055, pH = 1.26, pOH = 12.74, [OH-] = 1.8 x 10^-13.

Step 7: (vii) 0.055 M Ca(OH)2: [OH-] = 0.11, pOH = 0.96, pH = 13.04, [H+] = 9.1 x 10^-14.

Final answer: Table as above, using pH + pOH = 14 and [H+][OH-] = 10^-14.

Chapter 11: Organic Chemistry

Short Questions

1. What is meant by the term catenation? Give an example of a compound that displays catenation.

Catenation is the ability of an element to link with atoms of itself to form long chains or rings. Carbon shows catenation most strongly: for example, in butane (C4H10) four carbon atoms link together in a chain, and in benzene six carbon atoms link in a ring.

2. How is coal formed?

Coal is formed from dead plants buried under layers of earth millions of years ago. Under high pressure and temperature, and without air, the plant material slowly lost oxygen and hydrogen and became richer in carbon, passing through the stages of peat, lignite, bituminous coal and anthracite.

3. What is the importance of natural gas?

Natural gas is important as a clean fuel for cooking, heating, electricity generation and vehicles, and as a raw material for making fertilizers, plastics and other chemicals. It burns with a blue flame, produces less smoke than coal or oil, and is mainly methane.

4. Justify that organic compounds are used as food.

Organic compounds are used as food because our bodies need carbohydrates, proteins, fats and vitamins, all of which are organic

compounds. Carbohydrates and fats give energy, proteins build and repair body tissues, and vitamins regulate body functions, so food is essentially a mixture of organic compounds.

5. How are alkyl radicals formed? Explain with examples.

Alkyl radicals are formed when one hydrogen atom is removed from an alkane molecule. For example, removing one hydrogen from methane (CH4) gives the methyl radical (CH3-), and removing one from ethane (C2H6) gives the ethyl radical (C2H5-). They are represented by R- and are very reactive.

6. What is the difference between n-propyl and isopropyl radicals? Explain with structure.

n-Propyl (CH3-CH2-CH2-) is formed by removing a hydrogen from the end carbon of propane, so the free valency is on a primary carbon. Isopropyl (CH3-CH(-)-CH3) is formed by removing a hydrogen from the middle carbon, so the free valency is on a secondary carbon. They have the same formula C3H7 but different structures.

7. Explain different radicals of butane.

Butane (C4H10) gives two radicals: n-butyl (CH3-CH2-CH2-CH2-) by removing a hydrogen from an end carbon, and sec-butyl (CH3-CH2-CH(-)-CH3) by removing a hydrogen from a middle carbon. The two radicals differ in the position of the free valency and in their properties.

8. Define functional group with an example.

A functional group is an atom or a group of atoms that gives an organic

compound its characteristic properties and reactions. For example, the hydroxyl group (-OH) is the functional group of alcohols: ethanol (C_2H_5OH) behaves as an alcohol because of its -OH group.

9. What is an ester group? Write down the formula of ethyl acetate.

An ester group is -COO-, formed when an alcohol reacts with a carboxylic acid. Esters have sweet, fruity smells and are used as flavours and solvents. Ethyl acetate is $CH_3COOC_2H_5$, formed from acetic acid and ethanol.

10. Write down the dot and cross formulae of propane and n-butane.

Propane (C_3H_8): three carbon atoms in a chain, each carbon sharing electron pairs with hydrogen atoms and neighbouring carbons, giving 8 hydrogens in total (CH_3-CH_2-CH_3). n-Butane (C_4H_{10}): four carbon atoms in a straight chain with 10 hydrogens (CH_3-CH_2-CH_2-CH_3). In dot-and-cross diagrams, draw each shared pair as one dot from one atom and one cross from the other.

11. Define structural formula. Draw the structural formulae of n-butane and isobutane.

A structural formula shows how the atoms in a molecule are arranged and bonded to each other. n-Butane is CH_3-CH_2-CH_2-CH_3, a straight chain of four carbons. Isobutane is CH_3-$CH(-CH_3)$-CH_3, a branched chain in which the middle carbon carries an extra methyl group.

12. Write classification of coal.

Coal is classified into four types by carbon content: peat (about 60% carbon, soft and brown), lignite (about 70% carbon), bituminous coal

(about 80% carbon, the common coal used for fuel), and anthracite (about 90% carbon, hard, shiny and the best quality coal).

13. What are heterocyclic compounds? Give two examples.

Heterocyclic compounds are ring compounds in which the ring contains atoms other than carbon, such as nitrogen, oxygen or sulphur. Two examples are pyridine (a six-membered ring with one nitrogen atom) and furan (a five-membered ring with one oxygen atom).

14. Why are benzene and other homologous compounds of benzene called aromatic compounds?

Benzene and its homologues are called aromatic because they were first obtained from fragrant (aromatic) plant products such as balsams and resins, and because they share a special ring structure with delocalized electrons. They undergo substitution reactions rather than addition reactions, which is characteristic of aromatic compounds.

Extensive Questions

1. (a) How is coal formed? What are the different types of coal? (b) Write down the composition and uses of different types of coal.

(a) Coal formed from dead plants buried under earth over millions of years, under pressure and heat without air, changing from peat to lignite to bituminous coal to anthracite. (b) Peat (60% carbon) is used as a poor fuel; lignite (70%) is used in power stations; bituminous coal (80%) is the main fuel for industry and for making coke; anthracite (90%) is the cleanest, hardest coal, used for domestic heating and in metallurgy.

2. (a) What is destructive distillation of coal? (b) Name the different types of products obtained by the destructive distillation of coal.

(a) Destructive distillation of coal is heating coal strongly in the absence of air, so it decomposes instead of burning. (b) Its products are: coal gas (used as fuel), ammoniacal liquor (used to make fertilizers), coal tar (used for roads, dyes and medicines), and coke (a nearly pure carbon residue used as fuel and in iron extraction).

3. Write a detailed note on functional groups of alkenes and alkynes. How are they identified from other compounds?

The functional group of alkenes is the carbon-carbon double bond (C=C), and of alkynes it is the carbon-carbon triple bond (C≡C). They are identified by their addition reactions: both decolourize bromine water and dilute KMnO4 solution, which alkanes do not. Alkynes are also identified by their reaction with ammoniacal silver nitrate, which gives a white precipitate with terminal alkynes.

4. Give some uses of organic compounds in our daily life.

Organic compounds are used as food (carbohydrates, proteins, fats), clothing (cotton, wool, nylon), fuel (petrol, natural gas, kerosene), medicines (aspirin, antibiotics), soaps and detergents, plastics, paints, dyes, perfumes, and in the manufacture of paper, rubber and fertilizers.

5. Write down the characteristics of homologous series.

A homologous series is a family of compounds with the same functional group and general formula, differing by a -CH2- group between successive members. Its characteristics are: all members have similar chemical properties, physical properties change gradually with molecular

mass, all can be made by similar methods, and they have the same general formula, for example alkanes CnH2n+2.

6. Why are organic compounds numerous?

Organic compounds are numerous because carbon shows catenation, forming long chains, branches and rings of almost any length, and because carbon forms four strong covalent bonds, allowing many combinations with hydrogen, oxygen, nitrogen and halogens. Isomerism also multiplies the number: the same formula can give many different structures.

7. What are amines? Explain the different types of amines giving an example of each type. How is primary amino group identified?

Amines are organic compounds containing the amino group -NH2, derived from ammonia by replacing hydrogen atoms with alkyl groups. Primary amines have one alkyl group (methylamine, CH3NH2), secondary amines have two (dimethylamine), and tertiary amines have three (trimethylamine). A primary amino group is identified by the carbylamine test: warming with chloroform and alcoholic KOH gives a foul-smelling isocyanide.

8. Describe the functional group of an alcohol. How are alcoholic groups identified?

The functional group of an alcohol is the hydroxyl group -OH attached to a carbon atom. Alcohols are identified by their reaction with sodium metal, which gives hydrogen gas, and by the formation of a sweet-smelling ester when warmed with a carboxylic acid and concentrated sulphuric acid. Ethanol also gives a characteristic smell and

burns with a blue flame.

9. Differentiate between aldehydic and ketonic functional groups. How are both identified from each other?

The aldehydic group is -CHO, at the end of a carbon chain, while the ketonic group is >C=O, in the middle of a chain with two carbon groups attached. Aldehydes are identified by Tollens' test (silver mirror with ammoniacal silver nitrate) and Fehling's test (red precipitate), which ketones do not give, because aldehydes are easily oxidized while ketones are not.

10. Encircle the functional groups in the following compounds (compounds printed as figures in the book). Also give the names of the functional groups.

Identify each compound printed in the book and mark the atom or group that gives it its character: -OH is hydroxyl (alcohol), -CHO is aldehydic, >C=O is ketonic, -COOH is carboxylic acid, -COO- is ester, -NH2 is amino, and -X is halo group. Name the group found in each printed compound by matching it with this list.

11. What are the general properties of organic compounds?

Organic compounds are usually covalent, so they have low melting and boiling points, are often insoluble in water but soluble in organic solvents, do not conduct electricity, burn in air to give carbon dioxide and water, and react slowly compared with inorganic compounds. They often have characteristic smells and form a very large number of compounds.

12. Write a detailed note on classification of organic compounds.

Organic compounds are classified into: (i) open-chain (aliphatic) compounds, such as alkanes and alcohols; (ii) closed-chain (cyclic) compounds, divided into alicyclic compounds (cyclohexane) and aromatic compounds (benzene); and (iii) heterocyclic compounds, whose rings contain nitrogen, oxygen or sulphur (pyridine, furan). They are further classified by functional group into hydrocarbons, alcohols, aldehydes, ketones, acids, esters and amines.

Chapter 12: Hydrocarbons

Short Questions

1. Differentiate between saturated and unsaturated hydrocarbons.

Saturated hydrocarbons contain only single bonds between carbon atoms, so each carbon carries the maximum number of hydrogens, for example alkanes. Unsaturated hydrocarbons contain at least one double or triple bond, so they carry fewer hydrogens, for example alkenes and alkynes, and they undergo addition reactions.

2. A compound consisting of four carbon atoms has a triple bond in it. How many hydrogen atoms are present in it?

A four-carbon compound with one triple bond is an alkyne, general formula CnH2n-2. For n = 4, hydrogens = 2(4) - 2 = 6. The compound is butyne (C4H6).

3. Why are the alkanes called 'paraffins'?

Alkanes are called paraffins (from Latin parum affinis, meaning 'little affinity') because they are very unreactive. Their single bonds are strong and non-polar, so they do not readily react with acids, bases or oxidizing agents under ordinary conditions.

4. What do you know about hydrogenation of alkenes?

Hydrogenation of alkenes is the addition of hydrogen across the double bond in the presence of a nickel catalyst, converting an alkene into an alkane. For example, ethene + H2 → ethane. This reaction is used to convert liquid vegetable oils (unsaturated) into solid ghee (saturated).

5. How are alkyl halides reduced?

Alkyl halides are reduced by replacing the halogen atom with hydrogen. For example, methyl chloride reacts with zinc and acid, or with lithium aluminium hydride, to give methane: CH3Cl + [H] → CH4 + HCl. The halogen is removed as a halide ion.

6. Why are the alkanes used as fuel?

Alkanes are used as fuel because they burn in air with a clean flame, releasing a large amount of heat energy. Their combustion gives carbon dioxide and water, and fuels such as natural gas (methane), petrol and kerosene are all mixtures of alkanes.

7. How can you prepare ethene from alcohol and ethyl bromide?

From alcohol: ethanol is heated with concentrated sulphuric acid at 170 degrees C, which removes water (dehydration) to give ethene: C2H5OH → C2H4 + H2O. From ethyl bromide: ethyl bromide is heated with alcoholic potassium hydroxide, which removes HBr

(dehydrohalogenation) to give ethene: $C_2H_5Br + KOH \rightarrow C_2H_4 + KBr + H_2O$.

8. Identify propane from propene with a chemical test.

Pass each gas through bromine water: propene (an alkene) decolourizes bromine water because it adds bromine across its double bond, while propane (an alkane) does not change the colour. Propene also decolourizes dilute $KMnO_4$ solution, while propane does not.

9. Why are the alkenes called 'olefins'?

Alkenes are called olefins (oil-forming) because their lower members react with chlorine to form oily liquids. For example, ethene reacts with chlorine to give 1,2-dichloroethane, an oily liquid, and this oil-forming property gave the family its old name.

10. Why can't alkanes be oxidized with $KMnO_4$ solution?

Alkanes cannot be oxidized by $KMnO_4$ because they have only strong, non-polar single bonds and no double or triple bond for the oxidizing agent to attack. Dilute $KMnO_4$ therefore keeps its purple colour with alkanes, while it is decolourized by alkenes and alkynes.

11. What are the addition reactions? Explain with an example.

Addition reactions are reactions in which atoms or groups add across a double or triple bond, breaking the multiple bond and forming a saturated product. For example, ethene reacts with bromine: $C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$ (1,2-dibromoethane), where the double bond becomes a single bond and each carbon gains a bromine atom.

12. Justify that alkanes give substitution reactions.

Alkanes give substitution reactions because their single bonds cannot take part in addition, so a hydrogen atom is replaced by another atom or group instead. For example, methane reacts with chlorine in sunlight: $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$, where a hydrogen is substituted by chlorine. Further substitution gives CH_2Cl_2, $CHCl_3$ and CCl_4.

13. Both alkenes and alkynes are unsaturated hydrocarbons. State the one most significant difference between them.

The most significant difference is the type of multiple bond: alkenes contain a carbon-carbon double bond (C=C) and follow the general formula C_nH_{2n}, while alkynes contain a carbon-carbon triple bond (C≡C) and follow C_nH_{2n-2}. Alkynes are therefore more unsaturated and more reactive than alkenes.

14. Write the molecular, dot and cross and structural formula of ethyne.

Ethyne (acetylene) has molecular formula C_2H_2. Its structural formula is H-C≡C-H, with a triple bond between the two carbons. In dot-and-cross form, draw the two carbons sharing three pairs of electrons (the triple bond) and each carbon sharing one pair with a hydrogen atom.

15. Why are hydrocarbons soluble in organic solvents?

Hydrocarbons are non-polar covalent compounds, and they dissolve in non-polar organic solvents such as benzene, carbon tetrachloride and ether, following the rule 'like dissolves like'. They do not dissolve in water because water is polar and cannot break the weak forces between hydrocarbon molecules.

16. Give the physical properties of alkanes.

Alkanes are colourless, odourless (except the lower ones which have a faint smell), non-polar covalent compounds. Their melting and boiling points increase with molecular size. The first four are gases, the next ones up to about C17 are liquids, and higher ones are solids. They are insoluble in water, soluble in organic solvents, and less dense than water.

17. How can you identify ethane from ethene?

Ethene decolourizes bromine water and dilute KMnO4 solution, while ethane does not. Ethene also burns with a more smoky flame than ethane. The bromine water test is the standard identification: only the unsaturated ethene removes the reddish-brown colour.

18. Why does the colour of bromine water discharge on addition of ethene in it?

Ethene adds bromine across its double bond to form colourless 1,2-dibromoethane. The reddish-brown colour of bromine water is due to free bromine, and as bromine is consumed in the addition reaction, the colour disappears.

19. State one important use of each: (i) Ethene (ii) Acetylene (iii) Chloroform (iv) Carbon tetrachloride.

(i) Ethene: used to make polythene plastic and to ripen fruits artificially. (ii) Acetylene: used in oxy-acetylene welding and cutting of metals, and to make PVC. (iii) Chloroform: used as a solvent and formerly as an anaesthetic. (iv) Carbon tetrachloride: used as a solvent for fats and oils and in fire extinguishers (now limited because it harms the ozone layer).

Extensive Questions

1. What type of reactions are given by alkanes? Explain with reference to halogenation of alkanes.

Alkanes mainly give substitution reactions, in which a hydrogen atom is replaced by another atom, and combustion reactions. In halogenation, for example, methane reacts with chlorine in sunlight in stages: CH4 + Cl2 → CH3Cl + HCl, then CH3Cl + Cl2 → CH2Cl2 + HCl, and so on to CCl4. Each step substitutes one more hydrogen by chlorine, with sunlight providing the energy to break the chlorine molecule into reactive atoms.

2. Alkanes are a source of heat. Explain it.

Alkanes burn in excess oxygen to give carbon dioxide and water, releasing a large amount of heat energy because strong C=O and O-H bonds are formed. For example, CH4 + 2O2 → CO2 + 2H2O + heat. This is why natural gas, petrol, diesel and kerosene, all mixtures of alkanes, are used as fuels for cooking, heating, transport and power generation.

3. Prepare the following as directed: (i) ethane from ethene (ii) acetylene from tetrahalide (iii) carbon tetrachloride from methane (iv) ethylene glycol from ethene (v) 1,2-dibromoethane from ethene (vi) glyoxal from acetylene.

(i) Ethane from ethene: hydrogenation with H2 and nickel catalyst, C2H4 + H2 → C2H6. (ii) Acetylene from tetrahalide: 1,1,2,2-tetrabromoethane reacts with zinc dust (dehalogenation) to give acetylene. (iii) Carbon tetrachloride from methane: chlorination in sunlight through successive substitution to CCl4. (iv) Ethylene glycol from ethene: ethene is oxidized

with cold dilute KMnO4 (or via the epoxide) to give ethane-1,2-diol. (v) 1,2-dibromoethane from ethene: addition of bromine, C2H4 + Br2 → C2H4Br2. (vi) Glyoxal from acetylene: acetylene is oxidized (for example with nitric acid or by ozonolysis) to give glyoxal (OHC-CHO).

4. Explain the oxidation of acetylene.

Acetylene burns in air with a very hot, smoky flame: 2C2H2 + 5O2 → 4CO2 + 2H2O + heat. With a limited supply of oxygen it gives carbon monoxide and soot. Controlled oxidation with reagents such as KMnO4 or nitric acid breaks the triple bond and gives oxidized products such as glyoxal and then oxalic acid, showing that alkynes are easily oxidized because of their triple bond.

5. Write balanced chemical equations for the following reactions. Also, name the products that are formed. (i) A mixture of ethyne and hydrogen is passed over heated nickel (ii) Ethyne is treated with chlorine (iii) Ethyne is burnt in air (iv) Ethyne is passed through bromine water.

(i) C2H2 + 2H2 → C2H6 (ethane). (ii) C2H2 + 2Cl2 → C2H2Cl4 (1,1,2,2-tetrachloroethane). (iii) 2C2H2 + 5O2 → 4CO2 + 2H2O (carbon dioxide and water). (iv) C2H2 + 2Br2 → C2H2Br4 (1,1,2,2-tetrabromoethane); the bromine water is decolourized.

6. Explain briefly: (i) Why butane undergoes substitution reactions? (ii) There are millions of organic compounds. (iii) Acetylene undergoes addition reactions in two stages. (iv) Alkynes are more reactive than alkanes.

(i) Butane is an alkane with only single bonds, so it cannot add across a bond; instead a hydrogen is substituted by another atom, as in

chlorination. (ii) There are millions of organic compounds because carbon catenates into chains, branches and rings, forms bonds with many elements, and shows isomerism. (iii) Acetylene has a triple bond, so addition occurs in two stages: first one pi bond breaks to give an alkene, then the second pi bond breaks to give a saturated product. (iv) Alkynes are more reactive than alkanes because their triple bond has two easily broken pi bonds and a high electron density that attracts electrophiles.

Chapter 13: Biochemistry

Short Answer Questions

1. How do plants synthesize carbohydrates?

Plants synthesize carbohydrates by photosynthesis: in the presence of sunlight and chlorophyll, carbon dioxide and water combine to form glucose and oxygen: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$. The glucose is then converted into starch for storage and into cellulose for cell walls.

2. Give the characteristics of monosaccharides.

Monosaccharides are the simplest sugars, with one sugar unit that cannot be hydrolyzed further. They are sweet, soluble in water, crystalline solids, and have the general formula $(CH_2O)_n$. Examples are glucose, fructose and galactose, all with formula $C_6H_{12}O_6$ but different structures.

3. What is the difference between glucose and fructose?

Glucose and fructose are both monosaccharides with formula C6H12O6, but glucose is an aldose (it contains an aldehyde group -CHO) while fructose is a ketose (it contains a ketone group >C=O). Glucose is the main blood sugar and is less sweet, while fructose is the sweetest sugar, found in fruits and honey.

4. Give an example of a disaccharide. How is it hydrolyzed into monosaccharides?

Sucrose (table sugar) is a disaccharide made of glucose and fructose. On hydrolysis with dilute acid or the enzyme invertase, it splits into its two monosaccharides: sucrose + water → glucose + fructose. Similarly, maltose hydrolyzes into two glucose molecules and lactose into glucose and galactose.

5. Give the characteristics of polysaccharides.

Polysaccharides are large molecules made of many monosaccharide units joined together. They are usually not sweet, are insoluble or only slightly soluble in water, and do not taste sweet. Examples are starch (food storage in plants), glycogen (storage in animals) and cellulose (plant cell walls).

6. Where are the proteins found?

Proteins are found in all living cells and in foods such as meat, fish, eggs, milk, cheese, pulses, beans and nuts. In the body they form muscles, skin, hair, nails, enzymes, hormones and antibodies, and they are essential for growth and repair.

7. Describe the uses of carbohydrates.

Carbohydrates are the main source of energy for the body, as glucose is oxidized to release energy. They are also used as food (starch, sugar), as structural material (cellulose in plants), in making paper and cloth (cellulose), and as storage food (starch in plants, glycogen in animals).

8. Lactose is a disaccharide; which monosaccharides are present in it?

Lactose, the sugar of milk, is made of glucose and galactose joined together. On hydrolysis it splits into one molecule of glucose and one molecule of galactose.

9. Why are the ten amino acids essential for us?

Ten amino acids are called essential because the human body cannot make them, so they must be obtained from food. They are needed to build body proteins, enzymes and hormones, and without them growth, repair and normal body functions cannot continue.

10. How are proteins formed?

Proteins are formed when amino acids join together by peptide bonds, with the loss of a water molecule between the carboxyl group of one amino acid and the amino group of the next. Long chains of amino acids fold into specific shapes to form proteins, and the sequence of amino acids decides the protein's function.

11. How is gelatin obtained?

Gelatin is obtained by boiling animal skin, bones and connective tissue in water, which breaks down the protein collagen into gelatin. The solution is then filtered, concentrated and dried into sheets or powder, used in food, capsules and photography.

12. Give the general formula of the lipids.

Lipids are fats and oils, formed from fatty acids and glycerol. Their general composition is that of esters of glycerol with long-chain fatty acids; a fat molecule (triglyceride) is formed from one glycerol and three fatty acid molecules, with the loss of three water molecules.

13. Name two fatty acids with their formulae.

Two fatty acids are: palmitic acid, C15H31COOH (a saturated fatty acid found in palm oil), and oleic acid, C17H33COOH (an unsaturated fatty acid found in olive oil). Stearic acid, C17H35COOH, is another common example.

14. Give the types of vitamins.

Vitamins are divided into two types: fat-soluble vitamins (A, D, E and K), which are stored in body fat and liver, and water-soluble vitamins (B-complex and C), which dissolve in water, are not stored in large amounts, and must be taken regularly in food.

15. What is the significance of vitamins?

Vitamins are essential in small amounts for normal growth, health and resistance to disease. They regulate body processes, help enzymes work, protect against deficiency diseases (such as scurvy from lack of vitamin C and rickets from lack of vitamin D), and keep skin, eyes, bones and blood healthy.

16. Describe the sources and uses of vitamin A.

Vitamin A is found in fish liver oil, butter, eggs, milk, carrots, mangoes and green leafy vegetables (as carotene). It is needed for good eyesight,

healthy skin and mucous membranes, growth, and resistance to infections. Its deficiency causes night blindness and dry skin.

17. Justify that water soluble vitamins are not injurious to health.

Water-soluble vitamins (B-complex and C) are not injurious in normal amounts because any excess is dissolved in water and excreted in urine, so they do not build up to toxic levels in the body. Fat-soluble vitamins, by contrast, are stored in fat and can become harmful in excess.

18. What do you mean by genetic code of life?

The genetic code of life is the information stored in DNA that decides the characteristics of an organism. It is written in the sequence of four bases (adenine, thymine, guanine, cytosine) along the DNA molecule, and each set of three bases (a codon) specifies one amino acid in a protein, thus controlling heredity.

19. What is the function of DNA?

DNA stores and transmits hereditary information from parents to offspring. It carries the instructions for making all the proteins of the body, controls the growth and working of cells, and passes genetic characters from one generation to the next through its base sequence.

20. How do you justify that RNA works like a messenger?

RNA works like a messenger because it copies the genetic instructions from DNA in the nucleus and carries them to the ribosomes in the cytoplasm, where proteins are made. This messenger RNA (mRNA) tells the ribosome the exact sequence of amino acids to join, so the correct protein is built.

Long Answer Questions

1. What are carbohydrates? How are monosaccharides prepared? Give their characteristics.

Carbohydrates are organic compounds of carbon, hydrogen and oxygen, with hydrogen and oxygen in the ratio 2:1, first made by plants through photosynthesis. Monosaccharides are prepared by hydrolysis of disaccharides and polysaccharides: for example, starch hydrolyzes to glucose, and sucrose to glucose and fructose. Their characteristics are: they are sweet, soluble in water, crystalline, cannot be hydrolyzed further, and have formula $(CH_2O)_n$.

2. Explain oligosaccharides.

Oligosaccharides are carbohydrates made of a small number (2 to 10) of monosaccharide units joined by glycosidic bonds. The most important are disaccharides: sucrose (glucose + fructose), maltose (glucose + glucose) and lactose (glucose + galactose). They are sweet, soluble in water, and on hydrolysis they give their constituent monosaccharides.

3. What are polysaccharides? Give their properties.

Polysaccharides are large carbohydrates made of hundreds or thousands of monosaccharide units. Their properties are: they are not sweet, they are insoluble or sparingly soluble in water, they do not reduce Fehling's solution, and on complete hydrolysis they give monosaccharides. Examples are starch, glycogen and cellulose, which serve as food storage and structural material.

4. Explain the sources and uses of proteins.

Proteins are obtained from animal sources (meat, fish, eggs, milk) and plant sources (pulses, beans, nuts, wheat). Their uses in the body are: building and repairing tissues, forming enzymes that speed up reactions, forming hormones that regulate body functions, forming antibodies that fight disease, and providing energy when carbohydrates and fats are short.

5. Explain that amino acids are building blocks of proteins.

Amino acids are the building blocks of proteins because proteins are long chains of amino acids joined by peptide bonds. There are 20 common amino acids, and the order in which they are joined decides the shape and function of the protein. Just as letters form words, different sequences of amino acids form the thousands of different proteins in living things.

6. Explain the sources and uses of lipids.

Lipids come from animal fats (butter, ghee, meat fat) and plant oils (olive oil, sunflower oil, groundnut oil). Their uses are: they are a concentrated source of energy, they insulate the body against heat loss, they protect internal organs, they carry fat-soluble vitamins, and they form cell membranes.

7. Give the importance of vitamins.

Vitamins are important because, although needed in tiny amounts, they are essential for health: they prevent deficiency diseases, help the body use carbohydrates, proteins and fats, support growth, eyesight, blood formation and bone formation, and strengthen resistance to infections. A

diet lacking any vitamin leads to a specific deficiency disease.

8. Describe the sources, uses and deficiency symptoms of water soluble vitamins.

Water-soluble vitamins are the B-complex (B1, B2, B6, B12, niacin, folic acid) and vitamin C. Sources are whole grains, pulses, meat, eggs, milk, citrus fruits and green vegetables. They are used for energy release, blood formation, nerve function and wound healing. Deficiency symptoms are: beri-beri (B1), sore tongue and skin (B2), anaemia (B12, folic acid), pellagra (niacin), and scurvy with bleeding gums (vitamin C).

Chapter 14: Environmental Chemistry I - The Atmosphere

Short Answer Questions

1. Explain the phenomenon of decreasing temperature in troposphere.

In the troposphere, temperature decreases with height because the air is heated mainly from the Earth's surface, not directly by the Sun. The ground absorbs sunlight and warms the air touching it, and this warm air rises and cools as it expands. The higher air is therefore farther from the heat source and colder, falling by about 6.5 degrees C per kilometre.

2. Differentiate between primary and secondary air pollutants.

Primary pollutants are emitted directly into the air from a source, for example carbon monoxide, sulphur dioxide, nitrogen oxides and dust.

Secondary pollutants are not emitted directly but are formed in the air by reactions between primary pollutants, for example ozone and peroxyacetyl nitrate (PAN) formed by sunlight acting on nitrogen oxides and hydrocarbons in photochemical smog.

3. State the major sources of CO and CO2 emission.

Major sources of carbon monoxide are incomplete burning of fuels in vehicles, furnaces and cigarette smoke. Major sources of carbon dioxide are complete burning of fossil fuels (coal, oil, gas) in power stations, vehicles and factories, respiration of living things, and deforestation, which removes the trees that absorb CO2.

4. CO2 is responsible for heating up atmosphere, how?

Carbon dioxide is a greenhouse gas: it allows short-wave sunlight to pass through to the Earth but absorbs the long-wave infrared radiation given out by the warmed Earth, and re-radiates part of it back downwards. This traps heat in the atmosphere, warming it, just as glass traps heat in a greenhouse. Increasing CO2 therefore increases global warming.

5. CO is a hidden enemy, explain its action.

Carbon monoxide is a hidden enemy because it is colourless and odourless, so it cannot be seen or smelled, yet it is highly poisonous. It combines with haemoglobin in the blood about 300 times more strongly than oxygen, forming carboxyhaemoglobin, which cannot carry oxygen. The body's tissues are starved of oxygen, causing headache, dizziness, unconsciousness and death.

6. What threats are there to human health due to SO2 gas as air

pollutant?

Sulphur dioxide irritates the eyes, nose, throat and lungs, causing coughing, breathing difficulty and worsening of asthma and bronchitis. Long exposure damages lung tissue and reduces resistance to infections, and it also forms acid rain which harms crops, buildings and water life.

7. Which air pollutant is produced on anaerobic decomposition of organic matter?

Methane (CH4) is produced by anaerobic decomposition (decay without oxygen) of organic matter, for example in marshes, paddy fields, landfills and the stomachs of cattle. Hydrogen sulphide, with its rotten-egg smell, is also produced in smaller amounts during anaerobic decay.

8. How does acid rain increase the acidity of soil?

Acid rain contains sulphuric and nitric acids formed from SO2 and nitrogen oxides. When it falls on soil, its hydrogen ions replace useful nutrients such as calcium, magnesium and potassium on soil particles, washing them away. It also releases toxic aluminium ions into the soil solution, making the soil more acidic and harmful to plant roots.

9. Point out two serious effects of ozone depletion.

Two serious effects of ozone depletion are: (i) more ultraviolet radiation reaches the Earth, increasing skin cancer, eye cataracts and damage to the immune system in humans; (ii) it harms plants, plankton and aquatic life, reducing crop yields and disturbing food chains in the oceans.

10. How is ozone layer formed in stratosphere?

In the stratosphere, ultraviolet sunlight splits oxygen molecules (O2) into

oxygen atoms. Each free oxygen atom then combines with another oxygen molecule to form ozone (O3): O + O2 → O3. This ozone absorbs further ultraviolet radiation, which both protects life below and warms the stratosphere.

11. Why does 75% of the atmospheric mass lie within the troposphere?

About 75% of the atmosphere's mass lies in the troposphere because air is compressible and gravity pulls it downwards, so the lowest layer is the densest. The troposphere is also the layer in contact with the Earth's surface, where most water vapour and dust are present, adding to its mass.

12. How is ozone layer being depleted by chlorofluorocarbons?

Chlorofluorocarbons (CFCs), used in refrigerators and sprays, rise to the stratosphere where ultraviolet light breaks them apart, releasing chlorine atoms. Each chlorine atom destroys many ozone molecules in a chain reaction: Cl + O3 → ClO + O2, then ClO + O → Cl + O2, freeing the chlorine to destroy more ozone. This thins the ozone layer, especially over the poles.

Long Answer Questions

1. Write down the significance of atmospheric gases.

Atmospheric gases are significant because: nitrogen (78%) dilutes oxygen and is needed for proteins; oxygen (21%) is needed for respiration and combustion; carbon dioxide is needed for photosynthesis and keeps the Earth warm; argon is an inert filler; water vapour forms

clouds and rain; and ozone in the stratosphere absorbs harmful ultraviolet radiation. Together they make life on Earth possible.

2. Give the characteristics of troposphere. Why does temperature decrease upwards in this sphere?

The troposphere is the lowest layer, about 12 km thick, containing 75% of the air's mass and almost all water vapour and weather (clouds, rain, storms). Temperature decreases upwards because the air is heated from the ground: the Earth's surface absorbs sunlight and warms the air above it, and air higher up is farther from this heat source and expands and cools as pressure falls.

3. What are the characteristics of stratosphere? Why does temperature increase upwards in this sphere?

The stratosphere lies above the troposphere up to about 50 km, is dry and almost free of weather, and contains the ozone layer. Temperature increases upwards in the stratosphere because ozone absorbs ultraviolet radiation from the Sun, warming the upper part of the layer. The top of the stratosphere is therefore warmer than its bottom.

4. CO_2 is necessary for plants but why is its increasing concentration alarming for us?

Carbon dioxide is necessary for photosynthesis, but its increasing concentration is alarming because it is a greenhouse gas that traps heat and causes global warming. This leads to melting of glaciers and sea-level rise, changes in rainfall, droughts and floods, damage to crops and ecosystems, and more extreme weather, threatening human life and food supply.

5. Why is CO considered a health hazard?

Carbon monoxide is a health hazard because it is colourless and odourless, so it gives no warning, and it binds to haemoglobin about 300 times more strongly than oxygen. Even small amounts reduce the blood's oxygen-carrying capacity, causing headache, dizziness, tiredness and confusion, and larger amounts cause unconsciousness and death by suffocation.

6. Define acid rain. How is it formed and what are its effects?

Acid rain is rain with a pH below 5.6, made acidic by dissolved oxides of sulphur and nitrogen. It forms when SO2 and NOx from burning fossil fuels react with water and oxygen in the air to give sulphuric and nitric acids, which fall as rain or snow. Its effects are: damage to crops and forests, acidification of soil and lakes killing fish, corrosion of buildings and monuments (especially marble), and irritation of human lungs.

7. Compounds of sulphur are air pollutants. Describe the sources of these compounds along with their effects.

Sulphur compounds, mainly sulphur dioxide (SO2) and hydrogen sulphide (H2S), come from burning coal and oil, smelting of sulphide ores, volcanic eruptions, and decay of organic matter. Their effects are: irritation of eyes and lungs, worsening of asthma, formation of acid rain that damages soil, water life and buildings, and haziness (smog) that reduces visibility.

8. Where does ozone layer lie in atmosphere? How is it depleting and how can we prevent its depletion?

The ozone layer lies in the stratosphere, about 20 to 30 km above the

Earth. It is being depleted by chlorine and bromine released from chlorofluorocarbons (CFCs) and halons, which destroy ozone in chain reactions. Its depletion can be prevented by banning CFCs and halons (as under the Montreal Protocol), using safer substitutes in refrigerators and sprays, and reducing the use of chemicals that release chlorine into the atmosphere.

9. Oxides of nitrogen cause air pollution. Describe the sources of these compounds.

Oxides of nitrogen (NO and NO2) come mainly from vehicles and power stations, where high temperatures cause nitrogen and oxygen in the air to combine during fuel burning. Other sources are lightning, bacterial action in soil, and the use of nitrogen fertilizers. They cause photochemical smog, acid rain, irritation of lungs, and damage to plants.

Chapter 15: Water

Short Questions

1. How does water rise in plants?

Water rises in plants through the xylem vessels by a combination of forces: root pressure pushes water up from the roots, capillary action pulls it through the narrow vessels, and transpiration pull draws it upwards as water evaporates from the leaves. Cohesion between water molecules keeps the water column continuous.

2. Which forces are responsible for dissolving polar substances in water?

Polar substances dissolve in water because of the attraction between their charged or partly charged ends and the polar water molecules. Ion-dipole forces act between ions and water molecules, and hydrogen bonding acts between polar molecules and water. These forces pull the solute particles apart and surround them with water molecules (hydration).

3. Why are non-polar compounds insoluble in water?

Non-polar compounds are insoluble in water because water molecules are strongly attracted to each other by hydrogen bonds, and non-polar molecules cannot form such bonds with water. Water therefore prefers to stay bonded to itself rather than to surround the non-polar particles, so the non-polar substance remains separate.

4. How does water dissolve sugar and alcohol?

Sugar and alcohol dissolve in water because they contain hydroxyl (-OH) groups that form hydrogen bonds with water molecules. When sugar is added to water, its molecules are pulled apart by water molecules forming hydrogen bonds with them, and the sugar spreads evenly through the water. Alcohol mixes in the same way through its -OH group.

5. How does limestone dissolve in water?

Limestone (calcium carbonate) is almost insoluble in pure water, but it dissolves in water containing dissolved carbon dioxide, which forms weak carbonic acid. The acid reacts with calcium carbonate to give soluble calcium hydrogen carbonate: $CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$. This is how caves and underground water become hard.

6. Differentiate between soft and hard water.

Soft water lathers easily with soap and contains very few dissolved calcium and magnesium salts. Hard water does not lather easily with soap, forming a scum instead, because it contains dissolved calcium and magnesium hydrogen carbonates (temporary hardness) or sulphates and chlorides (permanent hardness).

7. What are the causes of hardness in water?

Hardness in water is caused by dissolved calcium and magnesium salts picked up as water passes over rocks and soil. Temporary hardness is caused by calcium and magnesium hydrogen carbonates, and permanent hardness by their sulphates and chlorides, which do not break down on boiling.

8. What are the effects of temporary hardness in water?

Temporary hardness wastes soap by forming scum, leaves deposits (fur) in kettles, boilers and pipes that waste fuel and can block pipes, makes water taste unpleasant, and makes washing and cooking less effective. It can, however, be removed by boiling, which precipitates the carbonates.

9. Mention the disadvantages of detergents.

Disadvantages of detergents are: they do not break down easily (non-biodegradable ones persist in water), they cause foaming in rivers and sewage works, their phosphates cause algal growth (eutrophication) that kills aquatic life, and they can irritate skin and harm water organisms.

10. What is the difference between biodegradable and

non-biodegradable substances?

Biodegradable substances can be broken down by bacteria and other micro-organisms into harmless substances, for example paper, food waste and soap. Non-biodegradable substances cannot be broken down by micro-organisms and persist in the environment for a long time, for example plastics, many detergents and polythene bags.

11. How do detergents make the water unfit for aquatic life?

Detergents make water unfit for aquatic life because their phosphates feed algal blooms that block sunlight and use up oxygen when they decay, suffocating fish. The detergent chemicals themselves are toxic to fish and other organisms, and the foam they form reduces oxygen exchange at the water surface.

12. Why are pesticides used?

Pesticides are used to kill pests such as insects, fungi, weeds and rodents that damage crops and spread disease. They protect food production, increase crop yields, and control disease-carrying insects such as mosquitoes, but their overuse pollutes water and harms useful organisms.

13. What are the reasons of waterborne diseases?

Waterborne diseases are caused by drinking water contaminated with bacteria, viruses and parasites from human and animal waste, sewage, industrial waste and agricultural runoff. Poor sanitation, broken sewers, flooding and lack of water treatment spread these germs into drinking water supplies.

14. How can waterborne diseases be prevented?

Waterborne diseases can be prevented by: treating drinking water by filtration, chlorination or boiling; proper disposal of sewage away from water sources; washing hands and keeping food clean; using clean, covered water storage; and vaccinating against diseases such as cholera and typhoid where available.

Extensive Questions

1. How does polarity of water molecule play its role to dissolve the substances?

The water molecule is polar because oxygen pulls electrons more strongly than hydrogen, giving oxygen a partial negative charge and hydrogens partial positive charges. This polarity lets water attract ions and polar molecules: the negative oxygen ends surround positive ions and the positive hydrogen ends surround negative ions, pulling solute particles apart and keeping them dissolved. Non-polar substances, which have no charges for water to attract, do not dissolve.

2. Explain the methods of removing permanent hardness.

Permanent hardness is removed by: (i) adding washing soda (sodium carbonate), which precipitates calcium and magnesium as insoluble carbonates that can be filtered off; (ii) the ion-exchange (permutit) method, in which hard water passes through a zeolite that exchanges its sodium ions for calcium and magnesium ions; and (iii) distillation, which leaves all dissolved salts behind. Boiling does not remove permanent hardness.

3. Explain the water pollution because of industrial waste.

Industrial waste pollutes water when factories discharge chemicals, dyes, heavy metals (lead, mercury, chromium), acids and hot water into rivers without treatment. These poisons kill fish and plants, make water unfit for drinking and irrigation, accumulate in food chains, and hot water reduces dissolved oxygen. Treatment and strict discharge limits are needed to control this pollution.

4. Justify the statement: household water is the reason of water pollution.

Household water causes pollution because sewage, detergents, soaps, oils and garbage from homes flow into drains and rivers. Detergent phosphates cause algal blooms, organic waste uses up oxygen as it decays, and germs from sewage spread waterborne diseases. If household wastewater is not treated before discharge, it becomes a major source of water pollution.

5. Explain that agricultural effluents are fatal for aquatic life.

Agricultural effluents (fertilizers, pesticides and animal waste washed from fields) are fatal for aquatic life because fertilizers cause algal blooms that block light and consume oxygen, pesticides poison fish, frogs and insects directly, and decaying organic waste further lowers dissolved oxygen. Fish suffocate, food chains collapse, and the water becomes unfit for other life.

6. Explain five important waterborne diseases. How can these be prevented?

Five waterborne diseases are: cholera (severe diarrhoea and dehydration), typhoid (fever and weakness), dysentery (bloody

diarrhoea), hepatitis A (liver infection and jaundice), and giardiasis (stomach cramps and diarrhoea). They are prevented by drinking boiled or chlorinated water, proper sewage disposal, hand washing, clean food handling, and vaccination against cholera and typhoid.

7. Give some disadvantages of hard water.

Disadvantages of hard water are: it wastes soap by forming scum, it forms scale in kettles, boilers and pipes that wastes fuel and blocks flow, it makes clothes stiff and dull after washing, it leaves spots on dishes and glass, and it can cause dryness of skin and hair. It is, however, not harmful to health in normal amounts.

8. What is water pollution? Describe the effects of using polluted water.

Water pollution is the contamination of water by harmful substances that make it unfit for its intended use. Effects of using polluted water are: waterborne diseases such as cholera, typhoid and dysentery; poisoning by heavy metals and chemicals; skin and stomach problems; death of fish and aquatic plants; damage to crops when used for irrigation; and foul taste and smell that make water unusable.

9. Explain the reasons, why water is considered a universal solvent.

Water is called a universal solvent because its polar molecules can dissolve a very large number of substances: ionic compounds (salts) by ion-dipole attraction, polar compounds (sugar, alcohol) by hydrogen bonding, and gases such as oxygen and carbon dioxide. Almost no other common liquid dissolves so many different substances, which is why water carries dissolved minerals, nutrients and wastes in nature and in living bodies.

10. Write a note on the treatment of sewage water.

Sewage treatment is carried out in stages: screening removes large objects; sedimentation settles out solids; biological treatment uses bacteria to break down organic matter in aeration tanks; and disinfection with chlorine kills germs. The treated water is then safe to release into rivers, and the sludge is dried and used as manure or disposed of safely.

Chapter 16: Chemical Industries

Short Questions

1. What role is played by pine oil in the froth flotation process?

In froth flotation, pine oil acts as a frothing agent: it is added to the powdered ore mixed with water, and when air is blown through, it helps to form a stable froth on the surface. The ore particles stick to the froth bubbles and rise to the top, while the useless rock (gangue) sinks, so the ore is separated and concentrated.

2. Name the various metallurgical operations.

The various metallurgical operations are: concentration of the ore (removal of gangue), roasting or calcination (heating to convert ore to oxide), smelting (reduction of the oxide with carbon or another reducing agent), and refining (purification of the metal, for example by electrolysis or bessemerization).

3. How is roasting carried out?

Roasting is carried out by heating a sulphide ore strongly in the presence of excess air, below its melting point. The sulphur is removed as sulphur dioxide gas and the ore is converted into its oxide, for example: $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$. The oxide is then easier to reduce to the metal.

4. Explain process of electrorefining.

In electrorefining, the impure metal is made the anode and a thin strip of pure metal the cathode, in a solution of a salt of that metal. When current passes, the anode dissolves and pure metal deposits on the cathode, while impurities either fall as anode mud or remain in solution. Copper is refined this way using copper sulphate solution.

5. What are the advantages of Solvay's process?

Advantages of Solvay's process for making washing soda are: the raw materials (common salt, limestone and ammonia) are cheap and abundant; ammonia is recovered and reused; the process gives a pure product; and it operates continuously on a large scale, making it economical for industry.

6. What is the principle of Solvay's process?

The principle of Solvay's process is that sodium hydrogen carbonate is much less soluble in cold water than other salts in the mixture, so it precipitates out. Ammoniacal brine (salt solution saturated with ammonia) is carbonated with CO_2, giving NH_4HCO_3, which reacts with NaCl to precipitate $NaHCO_3$, leaving ammonium chloride in solution. Heating $NaHCO_3$ gives washing soda.

7. What happens when ammonical brine is carbonated?

When ammoniacal brine (sodium chloride solution saturated with ammonia) is carbonated, carbon dioxide reacts with ammonia and water to form ammonium hydrogen carbonate, which then reacts with sodium chloride to give sodium hydrogen carbonate as a precipitate and ammonium chloride in solution: NaCl + NH3 + CO2 + H2O → NaHCO3 + NH4Cl.

8. How is NaHCO3 converted to Na2CO3?

Sodium hydrogen carbonate is converted to sodium carbonate by heating (calcination): 2NaHCO3 → Na2CO3 + CO2 + H2O. The carbon dioxide released is reused in the carbonation tower, and the sodium carbonate obtained is washing soda.

9. How is ammonia recovered in the Solvay's process?

Ammonia is recovered by treating the ammonium chloride left in solution with slaked lime (calcium hydroxide): 2NH4Cl + Ca(OH)2 → CaCl2 + 2NH3 + 2H2O. The ammonia gas released is collected and sent back to saturate fresh brine, so very little ammonia is consumed overall.

10. How is ammonia prepared for the synthesis of urea?

Ammonia for urea synthesis is prepared by Haber's process: nitrogen and hydrogen react at high pressure (about 200 atm) and temperature (about 450 degrees C) in the presence of an iron catalyst: N2 + 3H2 ⇌ 2NH3. The ammonia is then reacted with carbon dioxide to make urea: 2NH3 + CO2 → NH2CONH2 + H2O.

11. Describe the formation of petroleum.

Petroleum formed millions of years ago from the remains of tiny sea

plants and animals buried under layers of sand and mud. Under high pressure and temperature, and without air, the organic matter slowly changed into a dark, viscous liquid mixture of hydrocarbons, trapped in porous rocks beneath impermeable layers.

12. What is refining of petroleum and how is it carried out?

Refining of petroleum is the separation of crude oil into useful fractions by fractional distillation. Crude oil is heated to vapour in a furnace and passed into a tall fractionating tower, where vapours cool and condense at different heights according to their boiling points: gases at the top, then petrol, kerosene, diesel, fuel oil and bitumen at the bottom.

13. Give a use of kerosene oil.

Kerosene oil is used as a fuel for lamps and stoves in homes, as fuel for jet aircraft (aviation turbine fuel), and as a solvent for paints and insecticides.

14. Describe the difference between diesel oil and fuel oil.

Diesel oil is a lighter fraction (boiling range about 250-350 degrees C) used as fuel in diesel engines of trucks, buses and tractors. Fuel oil is a heavier, thicker fraction (boiling above 350 degrees C) used as fuel in furnaces, ships and power stations, and it needs preheating before it can flow and burn.

15. Write down the names of four fractions obtained by the fractional distillation of residual oil.

Four fractions obtained from residual oil are: lubricating oil (used for machinery), paraffin wax (used for candles and polishes),

bitumen/asphalt (used for road surfacing), and petroleum coke (used as fuel and in electrodes).

16. What is the difference between crude oil and residual oil?

Crude oil is the raw, unrefined mixture of hydrocarbons pumped from the ground, containing fractions from gases to heavy residues. Residual oil is what remains after the lighter fractions (petrol, kerosene, diesel) have been distilled off from crude oil; it is thick, dark and contains the heaviest hydrocarbons, used for fuel oil, lubricating oil and bitumen.

17. Which petroleum fraction is used in dry cleaning?

Petrol (gasoline) and the light solvent fraction known as petroleum ether / white spirit are used in dry cleaning, because they dissolve grease and dirt from clothes without using water. Carbon tetrachloride was also used, but it is now avoided because of its harmful effects.

Extensive Questions

1. Describe in detail the various processes involved in the concentration of ore. Explain your answer with the help of diagrams.

Concentration of ore (dressing) removes gangue by physical methods: (i) gravity separation, in which the heavier ore settles faster than gangue in water; (ii) magnetic separation, in which a magnetic ore is pulled away from non-magnetic gangue by a magnet; (iii) froth flotation, in which powdered ore with water and pine oil is agitated with air, so ore particles rise with the froth while gangue sinks; and (iv) leaching, in which the ore is dissolved in a suitable chemical and recovered from solution. Diagrams

show the settling tank, magnetic roller and flotation tank for each method.

2. Explain the process of roasting with reference to copper.

Roasting of copper ore (copper pyrites, CuFeS2) is heating the concentrated ore strongly in excess air. The sulphur is oxidized to sulphur dioxide and the iron and copper sulphides change into oxides and sulphates, giving a mixture of Cu2S and FeO. The iron oxide is then removed as slag with silica in smelting, leaving copper matte, which is further processed to blister copper.

3. Write a detailed note on Ammonia Solvay's process.

The Ammonia-Solvay process makes washing soda (Na2CO3) from common salt, limestone and ammonia. Brine is saturated with ammonia, then carbonated with CO2 from heated limestone to precipitate sodium hydrogen carbonate, which is filtered and heated to give washing soda. Ammonia is recovered from ammonium chloride with slaked lime and reused, and calcium chloride is the main by-product. The process is continuous, cheap and gives pure washing soda.

4. Write a note on fractional distillation of petroleum.

Fractional distillation separates crude oil into fractions by boiling point. Crude oil is heated to about 400 degrees C, vaporized, and passed into a fractionating tower with trays at decreasing temperatures upwards. Fractions condense at the level matching their boiling range: refinery gas at the top, then petrol, naphtha, kerosene, diesel, fuel oil, lubricating oil and wax, with bitumen remaining at the bottom. Each fraction is then further treated for use.

5. How is urea manufactured? Explain showing the flow sheet diagram.

Urea is manufactured by reacting ammonia with carbon dioxide under high pressure (about 200 atm) and temperature (about 190 degrees C): 2NH3 + CO2 → NH2COONH4 (ammonium carbamate), which then loses water to give urea: NH2COONH4 → NH2CONH2 + H2O. The flow sheet shows: ammonia and CO2 fed into a reactor, the product mixture passed to a separator where unreacted gases are recycled, and the urea solution concentrated, crystallized and granulated for use as fertilizer.

6. How is crude oil refined? Explain two important fractions of petroleum along with their usage.

Crude oil is refined by fractional distillation in a fractionating tower after heating to vapour, separating it by boiling point into fractions that are then purified. Two important fractions are: petrol (boiling range about 40-180 degrees C), used as fuel for cars and as a solvent; and diesel oil (about 250-350 degrees C), used as fuel for diesel engines in trucks, buses, tractors and generators.

7. Write a note in detail on smelting and bessemerization, giving a specific example.

Smelting is heating a roasted or calcined ore with a reducing agent (usually coke) and a flux in a furnace, so the metal oxide is reduced to molten metal and impurities form a slag that floats off. For example, in iron smelting, Fe2O3 + 3CO → 2Fe + 3CO2 in a blast furnace, with limestone flux removing silica as slag. Bessemerization is blowing hot air through molten metal (as in copper matte) to oxidize impurities: for copper, air oxidizes iron sulphide to iron oxide (removed as slag) and then copper sulphide to blister copper, giving a purer metal.